Pascal's triangle

1 The 4th row of Pascal's triangle consists of the numbers $1, 4, 6, 4, 1$.

a Write down the numbers in the 5th row of Pascal's triangle.

b Find the sum of the entries in the 5th row of Pascal's triangle.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1,\ 8,\ \square,\ 56,\ 70,\ \square,\ 28,\ \square,\ 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the expansion of $(x+y)^7$?

4 State the number of terms in the following expansions:

a $(4x+y)^6$ **b** $(m+y)^8$

5 Consider the expansion of $(y+k)^4$.

a How many terms will there be in the expansion?

b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following expansion:

$$(x+m)^4 = \square x^4m^0 + \square x^3m^1 + \square x^2m^2 + \square x^1m^3 + \square x^0m^4$$

Binomial theorem

7 Evaluate the following:

a $\binom{6}{5}$ **b** $\binom{14}{1}$ **c** $\binom{10}{9}$ **d** $\binom{8}{0}$

8 Consider the expansion of $(a-4)^5$. Will the 2nd term be positive or negative?

9 By considering $(x+y)^4 = (x+y)(x+y)(x+y)(x+y)$, in how many ways can three factors of x and one factor of y be chosen from the expansion to form the term which contains x^3y^1?

10 Consider the binomial $(4x+3y)^4$.

a State the first term in the expansion **b** State the last term in the expansion

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Find the specified term for the following expansions:

a Fifth term of $(b+2)^7$.

b Fifth term of $(c+d)^9$.

c Fifth term of $(3u-v)^6$.

d Tenth term of $\left(u+\frac{1}{2}\right)^{12}$.

e Seventh term of $\left(\frac{3y}{2}-\frac{2}{3y}\right)^{10}$.

f Eleventh term of $\left(2x+y^2\right)^{13}$.

g Sixteenth term of $\left(x-y^5\right)^{19}$.

h Middle term of $\left(4x^2+3y^3\right)^8$.

i Middle term of $\left(-3x^{-3}+2y^{-2}\right)^4$.

j Constant term of $\left(2y-\frac{1}{y^3}\right)^8$.

12 The first three terms of $(x+y)^{10}$ are x^{10}, $10x^9y$ and $45x^8y^2$. Using the symmetry of the coefficients, write the last three terms of the expansion.

13 In the expansion of $(m+k)^n$, the coefficient of the 9th term is 45.

a Find the value of n.

b Find the 9th term.

14 A particular term in the expansion of $\left(3a^2+\frac{p}{b}\right)^4$ is $\frac{96a^2}{b^3}$, for some constant p. Find the value of p.

15 Consider the expansion of $(a+b)^5$.

a Find the coefficient of the 3rd term.

b Which other term has the same coefficient as the 3rd term?

16 Which two terms in the expansion of $(u+v)^{11}$ have a coefficient of $\binom{11}{9}$?

17 Find the coefficient of x^{17} in the expansion of $\left(3x^2+2x\right)^{11}$.

18 Use the binomial theorem to expand the following expressions:

a $(1+x)^4$

b $(2+b)^5$

c $(p+q)^5$

d $(c-d)^8$

e $(u+4v)^6$

f $(y+3)^4$

g $(ka-b)^6$

h $(y-5)^3$

i $\left(y-\frac{1}{3}\right)^3$

j $\left(2+\frac{y}{2}\right)^3$

k $(3y+2r)^3$

l $\left(a-\frac{1}{a}\right)^3$

m $\left(u^2+3v^2\right)^3$

n $\left(1+\frac{4}{y}\right)^3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 Find:

a The term in $(3+\sqrt{x})^{10}$ that contains x^4.

b The term in $(c+d)^{10}$ that contains c^3.

c The term in $(u+v)^{11}$ that contains v^8.

d The term in $(u+v)^{15}$ that contains u^5.

20 If the eighth and tenth terms in the expansion of $(x+y)^n$ have the same coefficients, find the value of n.

21 Find the coefficient of a^3b^6 in the expansion of $\left(2a-\frac{b}{2}\right)^9$.

22 Consider the expansion of $(1+x)^9$. State the coefficient of the 4th term.

23 Find the value of n such that the expansion of $(a+b)^n$ contains the term $84a^6b^3$.

24 Consider the binomial series:

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots$$

a Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.

b Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.

c Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647